

TVD Verification Report

UrbanEV–Maryland validation statistics: independent recomputation

Verification of all total-variation-distance values quoted in the TRB manuscript

July 24, 2026

Method

Every TVD quoted in the manuscript was recomputed from the underlying data (`pipeline/data/interim` parquet files, ACS 2020–2024 tables, and the converged baseline outputs), using the standard definition

$$\text{TVD}(p, q) = \frac{1}{2} \sum_k |p_k - q_k|,$$

computed on normalized category shares. Universe and weight handling follows the figure scripts: person-universe attributes (age, gender, employment) use person weights (`wtperfin`); household-universe attributes (income, size, vehicles, workers, tenure, dwelling) use household weights (`wthhfin`), with the synthetic person frame reweighted by $1/\text{hhsiz}$ to form a household frame. The survey comparison uses the held-out TEST split only (never seen in training); the ACS comparison is fully independent. Trip TVDs are computed on binned distributions (log-spaced distance bins, hourly departure bins, daily-VMT bins) against the weighted held-out survey (`wttrdfin`).

Population marginals: synthetic vs. held-out survey

Attribute	TVD
Age band	0.056
Gender	0.034
Employment status	0.079
Household income	0.040
Household size	0.046
Vehicles	0.039
Workers	0.045
Tenure	0.013
Dwelling type	0.056
Mean	0.0453
Quoted in manuscript	0.045 ✓

Population marginals: synthetic vs. independent ACS 2020–2024

Attribute	TVD
Age band	0.061
Gender	0.027
Household income	0.142
Household size	0.051
Vehicles	0.076
Tenure	0.074
Dwelling type	0.062
Mean	0.0703
Quoted in manuscript	0.070 ✓

Employment and workers have no matching ACS table and are excluded from the ACS mean, as stated in the figure.

Trip marginals: synthetic vs. held-out survey

Distribution	Recomputed	Quoted
Trip distance	0.046	0.046 ✓
Departure hour	0.036	0.036 ✓
Daily VMT per driver	0.063	0.063 ✓

Joint structure

The bias-corrected Cramér’s V association matrices (survey vs. synthetic) differ by a mean absolute error of 0.057, matching the quoted value. ✓

Observations

1. Income vs. ACS is 0.142, the only per-attribute value above the 0.10 benchmark. It is disclosed in the figure panel itself. The likely cause is temporal drift, not model error: the survey is 2017–18 and the ACS window is 2020–24, over which nominal household incomes rose substantially; against its own held-out survey the income TVD is 0.040.
2. One rounding inconsistency was found and fixed: the thesis scorecard figure displayed the ACS mean as 0.069; the recomputed value is 0.0703, which rounds to 0.070 as the manuscript states. The scorecard was regenerated with 0.070.

Conclusion

All TVD values quoted in the manuscript (0.045 survey mean, 0.070 ACS mean, 0.046/0.036/0.063 trip distributions) and the Cramér’s V error (0.057) reproduce exactly from the underlying data under the standard TVD definition with correct weighting and universe handling.